

Module 13)>>4. Conditional Statements

- **Introduction to conditional statements: if, else, elif**

1. Basic if Statement

- Executes a block of code only if a condition is True.
- Syntax:

if condition:

Code to execute if condition is True

2. if-else Statement

- Executes one block if the condition is True, and another if False.
- Syntax:

if condition:

Code if True

else:

Code if False

3. if-elif-else (Multiple Conditions)

- Checks multiple conditions in order.
- Stops at the first True condition and executes its block.
- Syntax:

if condition1:

Code if condition1 is True

elif condition2:

Code if condition2 is True

else:

Code if all conditions are False

- **Nested if-else conditions**

Nested if-else statements allow you to check multiple conditions in a hierarchical manner. This is useful when decisions depend on more than one condition.

1. Basic Structure of Nested if-else

- An if-else block inside another if-else block.
- Syntax:

```
if condition1:
```

```
    if condition2:
```

```
        # Code if condition1 and condition2 are True
```

```
    else:
```

```
        # Code if condition1 is True but condition2 is False
```

```
else:
```

```
    # Code if condition1 is False
```